

Simulation and Analysis of Synchronous Machines

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Abstract—This paper presents the simulation and analysis of synchronous machines under several operating conditions. Three studies are carried out in Simulink/Matlab: the effect of load changes on a synchronous motor, the effect of field-current variation on a synchronous motor, and the action of a governor on a synchronous generator during a switching event. The models are complemented with a pi line model and a three-phase breaker. The first study addresses the motor response to mechanical load changes in terms of delivered power and load angle. The second examines the motor under field-voltage variation, determining the resulting power angle and power factor. The third incorporates a governor and a pi line model to analyze the mechanical power output, the excitation system response, and other critical generator parameters.

Index Terms—Synchronous motor, synchronous generator, load change, field-current change, governor, Simulink–Matlab.

I. INTRODUCTION

A synchronous machine is a rotating AC electrical machine in which the shaft rotational speed and the electrical frequency are synchronized and mutually dependent. As studied throughout the Electrical Machines II course, the machine can operate either as a motor or as a generator. It is used mainly as an AC generator and, to a lesser extent, as an AC motor, since it does not develop starting torque on its own and therefore requires specific starting and acceleration methods to reach synchronous speed.

Through a set of simulations, together with theoretical analysis and observation, this work highlights key aspects of the synchronous machine and evaluates its behavior under different conditions. The aim is to understand its industrial use and applications, as well as how these machines relate to other engineering fields such as control and power generation.

II. THEORETICAL BACKGROUND

A. Starting of Synchronous Machines

To understand the nature of the starting problem, consider the instant at which power is applied to the stator windings. The rotor is stationary, and so is its magnetic field, while the stator magnetic field begins to rotate at synchronous speed. Synchronous operation is reached only when the rotor and stator magnetic fields are aligned. Three methods can be used to start the machine safely [1].

1) *Starting by reducing the electrical frequency:* If the stator magnetic field rotates slowly enough, the rotor can accelerate and lock to it; the stator field speed is then raised gradually by increasing f_e up to its rated value. Solid-state motor drives can convert a constant input frequency into any desired output frequency. When such a variable-frequency unit is included in the motor control circuit, starting is straightforward: the frequency is set to a very low value for starting and then raised to the rated operating frequency.

2) *Starting with an external prime mover:* An external starting motor brings the synchronous machine up to full speed. The synchronous machine is then connected in parallel with the power system as a generator, and the prime mover is disconnected from the shaft. Once the starting motor is removed, the shaft slows down, the rotor magnetic field falls behind, and the machine begins to behave as a motor; the synchronous motor can then be loaded normally. Because the starting motor only has to overcome the inertia of the unloaded synchronous machine, it can have a much smaller rating than the machine it starts.

3) *Starting with damper (amortisseur) windings:* This is one of the most common methods. At synchronous speed the damper windings have no steady-state effect, since no currents are induced in them. The machine is first started as an induction machine; when a speed close to synchronous is reached, DC is applied to the field winding, producing speed oscillations about synchronous speed that settle within a few cycles until rated speed is reached. This process is known as self-synchronization. Once started, the field current can be regulated so that the machine operates under- or over-excited to adjust its power factor; the machine can thus perform the dual role of driving a mechanical load and compensating reactive current on the network.

B. Effect of Load Changes on Synchronous Motors

For a fixed shaft load, a synchronous motor develops enough torque to remain rotating at synchronous speed together with its load [1]. If the shaft load increases, the rotor initially slows down; the torque angle δ becomes larger and the induced torque rises. The increased induced torque eventually accelerates the rotor back to synchronous speed, but with a larger torque angle δ .

The phasor diagram changes during this process according to the constraints on the machine. The internal generated voltage $E_A = K\phi\omega$ depends only on field current and speed.

Speed is held constant by the input power source, and since the field circuit is unchanged, the field current is also constant; therefore E_A must remain constant when the load changes. This behavior is illustrated in Fig. 1.

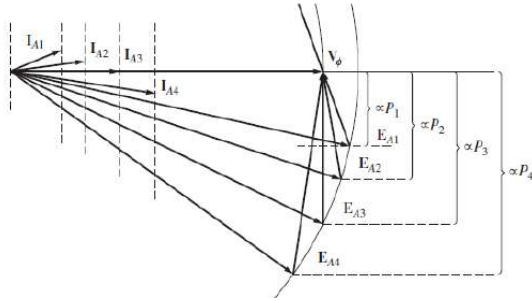


Fig. 1. Effect of increasing the load on the operation of a synchronous motor

C. Effect of Field-Current Changes on Synchronous Motors

A change in field current changes the magnitude of E_A but does not change the real power delivered by the motor. The real power changes only when the shaft load torque changes; since field current affects neither shaft speed nor shaft load, the real power is unaffected [1]. Consequently, the distances proportional to power in the phasor diagram remain constant: as the field current increases, E_A increases by sliding along the constant-power line, as shown in Fig. 2.

As E_A increases, the armature current first decreases and then increases. At low E_A the armature current lags and the motor behaves as an inductive load; as the field current rises, the armature current aligns with the load and the motor becomes resistive; with further increase the armature current leads and the motor becomes a capacitive load. Therefore, by controlling the field current of a synchronous motor, the reactive power supplied to or absorbed from the power system can be controlled.

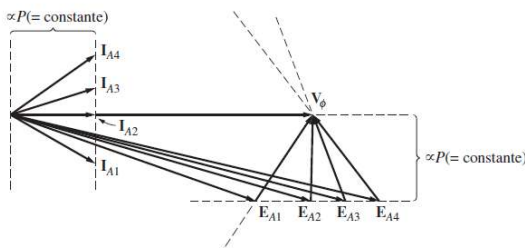


Fig. 2. Effect of increasing the field current on the operation of the motor

When E_A is smaller than V_ϕ , the motor draws a lagging current and absorbs reactive power (under-excited). When E_A is larger than V_ϕ , the motor draws a leading current and supplies reactive power (over-excited). The variation of the power factor with field current is shown in Fig. 3.

On each curve the minimum armature current corresponds to unity power factor; at any other point the machine exchanges reactive power. When the field current is below the

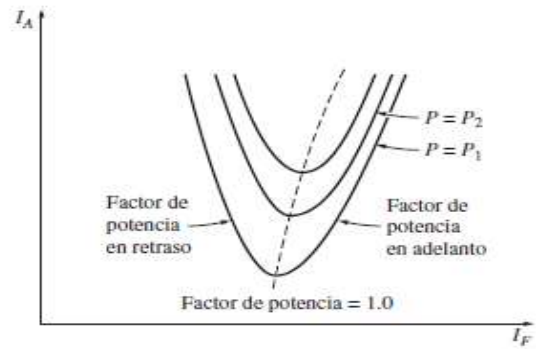


Fig. 3. Power-factor relationship in a synchronous motor

unity-power-factor point, the armature current lags and the motor absorbs reactive power; when it is above that point, the armature current leads and the motor supplies reactive power.

D. Governor of a Synchronous Generator

A governor (speed limiter or controller) measures and regulates the speed of a machine. It acts on the valve position to control the shaft speed in rpm, so that a desired speed can be selected through a control input. It automatically keeps the rotational speed of a prime mover within limits, independently of the load, typically by varying the rate at which fuel or working fluid is supplied. Most governors rely on centrifugal force: a pair of masses rotates around a spindle driven by the prime mover and is held in by a control force, usually applied by springs. As speed rises, the control force is overcome and the masses move outward; their motion is transmitted to the valves that feed the prime mover [2].

E. Automatic Voltage Regulator (AVR)

The automatic voltage regulator monitors the input and output voltage and regulates itself to keep the output voltage at a constant level automatically [3]. AVRs must allow remote adjustment of the generator output voltage and must sense the generated reactive power through a transformer, producing a proportional voltage drop. Their key requirements are a fast response to system events and full integration with the power system to keep the voltage within the machine's rated limits.

The AVR is essentially a controller: when the output voltage tends to differ from the reference, it increases or decreases the excitation voltage until the optimal output is reached. Because changing the rotor magnetic field and reading the result at the alternator output takes a finite time, the correction can become excessive and trigger an opposite, also-delayed correction, leading to a pendular (oscillatory) behavior over time. To prevent this, a filtering or delay mechanism, called a stabilizer, is included in all AVRs.

F. PI Line Model

The pi model is a geometric representation consisting of a series resistance and inductance placed between two shunt admittances that account for the line capacitance. This representation is particularly effective for very short transmission

lines, both overhead and underground, and can be treated as a cascade of sections, which simplifies the analysis of the system characteristics [4].

G. Machine Droop

Droop refers to the change in angular speed of the machine as it goes from no-load to full-load operation. If generation exceeds demand, there is a speed surplus and therefore a frequency increase; conversely, if demand exceeds generation, a drop in speed and frequency occurs [5]. Fig. 4 shows the relationship between power and frequency in a synchronous machine: as the load percentage increases, the speed—and therefore the frequency—decreases.

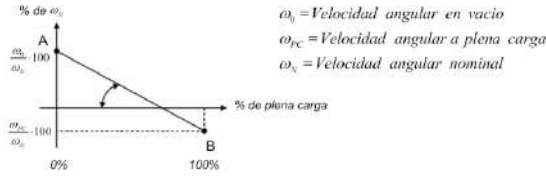


Fig. 4. Droop in a synchronous machine

H. Primary and Secondary Frequency Regulation

Primary regulation. Primary frequency regulation (PFR) balances the generated power against demand when the system operates under normal conditions. Its speed of action is limited by the inertia of the machines and is implemented through equipment that automatically modifies their output [6], [7]. In essence, PFR is intended to instantaneously balance generation with demand while keeping the frequency deviation to a minimum.

Secondary regulation. Secondary frequency regulation (SFR) is the manual or automatic action on the load setpoints of a dedicated group of machines, compensating the residual frequency error left by PFR. Its main function is to absorb demand variations with respect to the forecast during normal operation [6], [7]. SFR returns the machines to their assigned nominal values; its response time is several minutes, the time needed to optimally reallocate generation resources to meet demand.

III. METHODOLOGY AND CALCULATIONS

A. Simulation of Load-Change Effects on a Synchronous Motor

This simulation analyzes the behavior of a synchronous motor under changes in the mechanical load. The motor parameters are a line voltage of 762.1 V_{RMS} and an excitation voltage of 17.888 V. The motor initially delivers 48 864 W at a load angle of 21.13°. After 0.2 s the torque is halved and the new power angle is required; after 1 s the load angle is 26.27° and the corresponding power is required.

The fundamental expression for synchronous motors is given in (1).

$$\tau_{ind} = \frac{3 V_{\phi} E_A}{x_s \omega_s} \sin \delta \quad (1)$$

Replacing τ_{ind}/ω_s by the power yields (2).

$$P = \frac{3 V_{\phi} E_A}{x_s} \sin \delta \quad (2)$$

When the load changes, speed remains constant and, since the field circuit is untouched, the back-EMF is constant, as expressed in (3).

$$E_{A1} = E_{A2} = E_{An} \quad (3)$$

Using (2) for two load states and eliminating the common terms with (3) gives (4), from which the new load angle is obtained directly.

$$\delta_2 = \sin^{-1} \left(\frac{P_2 \sin \delta_1}{P_1} \right) \quad (4)$$

$$\delta_2 = \sin^{-1} \left(\frac{24\,432 \sin 21.13^\circ}{48\,864} \right) = 10.384^\circ$$

Thus, halving the torque—and therefore the power—reduces the load angle. For the third state, the power is the unknown; applying the same reasoning yields (5).

$$P_3 = \frac{P_2 \sin \delta_3}{\sin \delta_2} \quad (5)$$

$$P_3 = \frac{24\,432 \sin 26.27^\circ}{\sin 10.384^\circ} = 59\,994.94 \text{ W}$$

The simulation must verify whether this power produces a load angle equal (or close) to the value required for the third state.

B. Simulation of Field-Current-Change Effects on a Synchronous Motor

This simulation analyzes the synchronous motor under a change in field current—i.e., a variation in the back-EMF—which is the complementary case to the previous one. Initially the field voltage is 17.888 V at a power of 60 000 W. The field voltage is then reduced by 3 % and the new power angle is required, given that the first one is 26.46°.

When the field is varied, the power remains constant and only the voltage and the load angle change; the field change is also related to the power factor. Since the back-EMF is proportional to the field voltage, the constant-power constraint applied to Fig. 2 yields (6).

$$E_{A1} \sin \delta_1 = E_{A2} \sin \delta_2 \quad (6)$$

Solving for the load angle in the second state gives (7), from which the result follows directly.

$$\delta_2 = \sin^{-1} \left(\frac{E_{A1} \sin \delta_1}{E_{A2}} \right) \quad (7)$$

$$\delta_2 = \sin^{-1} \left(\frac{17.888 \sin 26.46^\circ}{17.35136} \right) = 27.345^\circ$$

C. Simulation of Governor Action During a Switching Event

No calculation is required for this simulation, since the model was provided by the instructor. This part of the practice consists of analyzing the effect of a synchronous generator in the proposed system and observing its response to a momentary loss of load. The following section analyzes the results of each simulation, with most emphasis placed on this one, since the first two are essentially verifications of theoretical results.

IV. SIMULATION RESULTS AND DISCUSSION

A. Load Changes in Synchronous Motors

This case analyzes the motor under a load change through three scenarios: the nominal state, a load reduction, and a load increase. Table I summarizes the theoretical results, the most important values being the load angle and the output power.

TABLE I
THEORETICAL SYSTEM VALUES, CASE 1.

Scenario	Load angle [δ]	Output power [W]
Scenario 1	21.130°	-48 864
Scenario 2	10.384°	-24 432
Scenario 3	26.270°	-59 994.94

The simulation must confirm whether, in scenario 2, the given power produces the calculated load angle and, in scenario 3, whether the given load angle produces the calculated power.

1) *Load Reduction to Half:* Here the load is halved at 0.2 s. Fig. 5 shows the load-angle signal. From $t = 0$ the load angle is -21.16° , close to the value stated in the exercise. At $t = 0.2$ s the angle starts to adjust to the new load and, from about 3 s onward, it settles at -10.43° , very close to the value in Table I; the error between simulated and calculated results is therefore small.

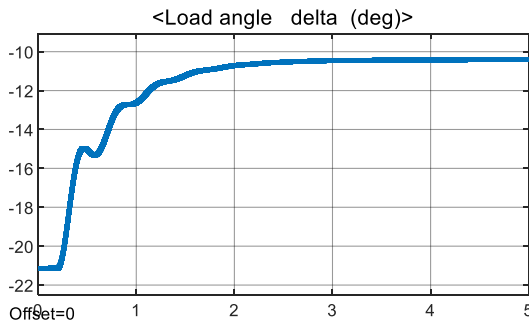


Fig. 5. Load angle for a load reduction to half, scenario 2, case 1

The motor power and speed are shown in Fig. 6 and Fig. 7.

Both signals change abruptly at the load change and then settle at their new values. The power starts near 48 kW and, after 0.2 s, settles at about 24 kW, matching the simulated load change and confirming that the power update is correct. The speed starts at about 1800 rpm, adjusts transiently after 0.2 s, and then returns to 1800 rpm.

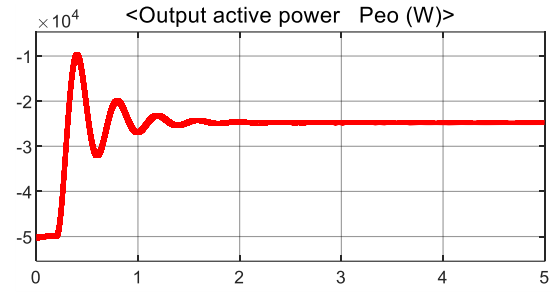


Fig. 6. Power delivered by the motor under a load reduction, scenario 2, case 1

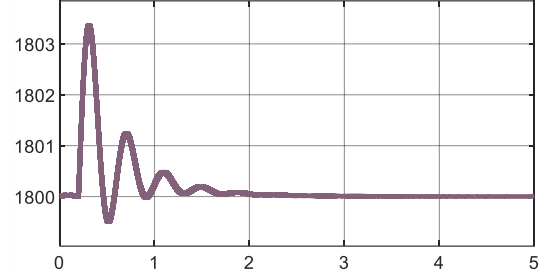


Fig. 7. Motor speed under a load reduction, scenario 2, case 1

This confirms the expected behavior: under a load change the motor speed fluctuates transiently, but the machine adjusts its torque to drive the new load at the same speed. In other words, the transient speed variation is compensated by a torque adjustment. This is verified in Fig. 8: when the load decreases, the speed tends to rise and, to counteract it, the torque is reduced.

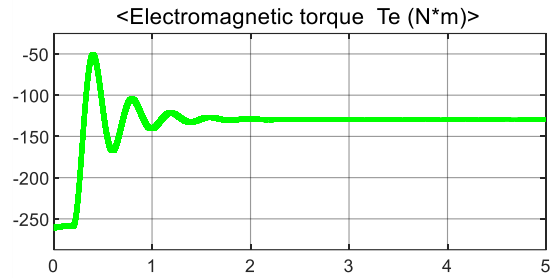


Fig. 8. Machine torque variation under a load reduction, scenario 2, case 1

2) *Load Increase:* Here the load is changed at $t = 1$ s. The calculation predicts an output power of 59 994.94 W (about 60 kW) after 1 s: starting from 21.13° at 48 864 W, the angle must reach 26.27° for the higher power. Fig. 9 shows the load angle starting at -21.16° and, after a transient, settling at -26.45° , close to the value in Table I and confirming the calculations with minimal error.

Fig. 10 and Fig. 11 show the power and speed, which now increase from $t = 1$ s.

The same principle holds: the motor keeps a constant speed for the new power. Because the load now increases, the speed drops momentarily and then, with the torque adjustment,

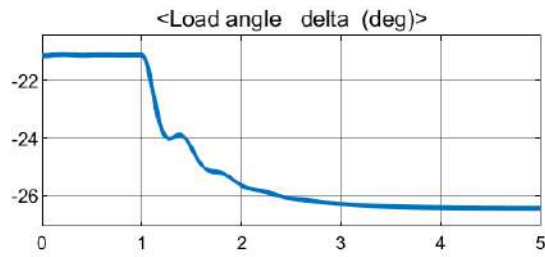


Fig. 9. Load angle for a load increase, scenario 3, case 1



Fig. 10. Power delivered by the motor under a load increase, scenario 3, case 1

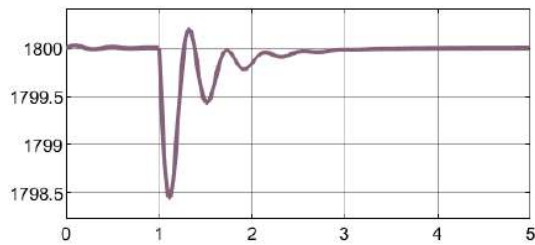


Fig. 11. Motor speed under a load increase, scenario 3, case 1

settles again at 1800 rpm. As the load increases, the speed tends to fall, so the torque must increase to prevent it, as shown in Fig. 12.

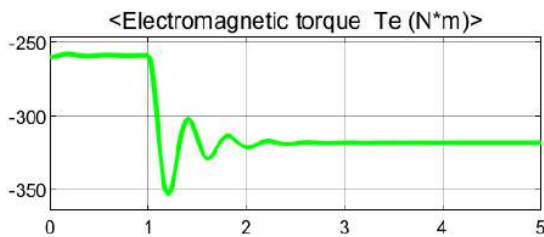


Fig. 12. Machine torque variation under a load increase, scenario 3, case 1

B. Field-Current Changes in a Synchronous Motor

The procedure is similar to the previous one, except that the power is now constant and the field current varies. Varying the field current changes the magnetic flux and the back-EMF; keeping the power fixed allows the power factor to be varied. The result to be verified is the load angle when the field is reduced, calculated as 27.345° .

From the theory, reducing the field current (or the back-EMF E_A) increases the load angle. Moreover, from Fig. 3, when the field current falls below its initial value the power factor also decreases and may enter a lagging (inductive) regime. The simulation is divided into a nominal scenario and a scenario in which the field voltage falls by 3 %.

1) *Nominal Operating Condition:* In the nominal scenario the load angle should settle at 26.46° and the power at 60 000 W. Fig. 13 shows the load angle settling at 26.46° , matching the value proposed in the exercise.

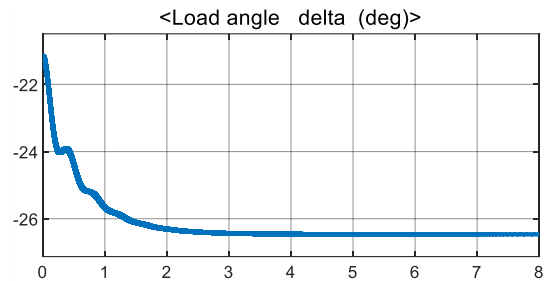


Fig. 13. Load angle for the nominal scenario, exercise 2

The power factor, obtained trigonometrically from the measured output active and reactive power, is shown in Fig. 14. It settles at 0.999, indicating a highly efficient power factor.

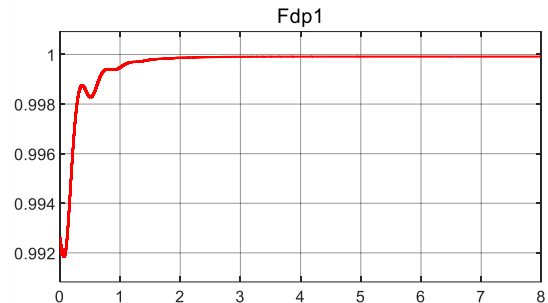


Fig. 14. Power factor for the nominal scenario, exercise 2

2) *Field Voltage Reduction:* This scenario verifies the new load angle, calculated as 27.345° , and observes the power factor. Fig. 15 shows the load angle settling at 27.16° , very close to the calculated value and confirming the hypothesis that reducing the field current increases the load angle.

The power factor in this scenario, shown in Fig. 16, settles at 0.9987.

The two power factors appear almost identical; the combined scenario is analyzed next to better quantify the difference produced by the field-voltage change. For now, the calculation of the new load angle under a reduced field voltage is confirmed.

3) *Combined Scenario:* This scenario joins the two previous ones: the field voltage falls by 3 % at $t = 3$ s. Fig. 17 shows that the settling time is sufficient for the first stage to reach 26.43° ; after 3 s the load angle moves to a new value of 27.16° .

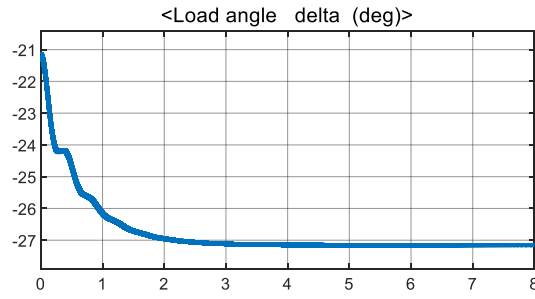


Fig. 15. Load angle for the field-voltage-reduction scenario, exercise 2

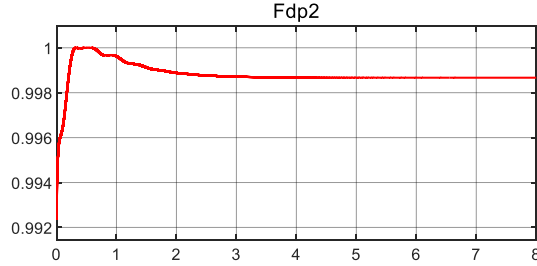


Fig. 16. Power factor for the field-voltage-reduction scenario, exercise 2

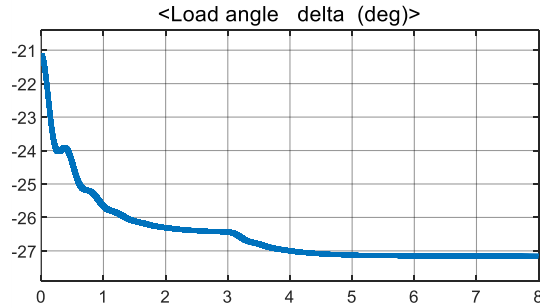


Fig. 17. Load angle for the combined scenario, exercise 2

The combined scenario yields values close to the individual analyses and shows the angle change over time more clearly. The power factor, shown in Fig. 18, drops noticeably after the first 3 s.

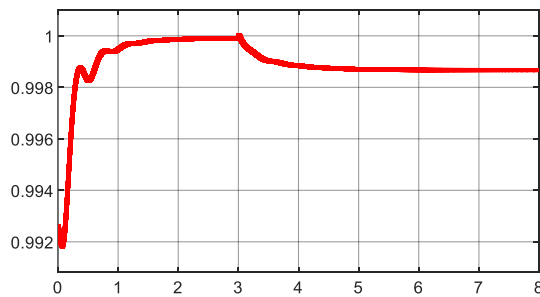


Fig. 18. Power factor for the combined scenario, exercise 2

Reducing the field voltage therefore also reduces the power factor. Here the field-voltage drop was only 3 %, so the de-

crease is acceptable; a larger reduction—for example 30 %—yields a power factor of 0.8302, showing that the power factor of a synchronous motor depends directly on the field voltage. The opposite occurs when the field voltage increases: with constant power, an increase in back-EMF raises the power factor and moves the machine into a capacitive (leading) regime.

The reactive-power plots confirm this transition: in the nominal scenario the reactive power is positive (inductive), whereas in the reduced-field scenario it becomes negative (capacitive), as shown in Fig. 19.

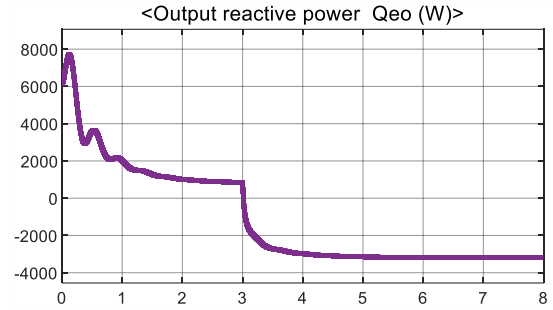


Fig. 19. Reactive power in the combined scenario, exercise 2

Finally, the motor speed under this change is shown in Fig. 20. At $t = 3$ s the speed dips briefly and then settles again, preserving the defining feature of synchronous motors: constant operating speed, with any transient deviation compensated.

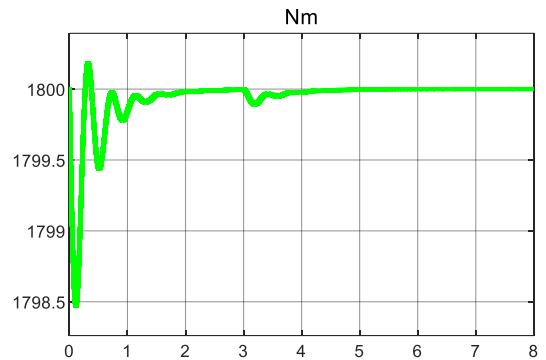


Fig. 20. Motor speed in the combined scenario, exercise 2

C. Governor Action

This simulation is purely an analysis of the proposed system; no calculations are verified. The objective is to observe the system response to a momentary loss of load. A governor block and an exciter block provide the mechanical power P_m and the field voltage V_f required by the machine. These blocks are fed back with the generator information throughout the simulation. During the first 2.5 s the system is stable with only small variations. When the breaker opens, the transformer, the largest load, and the generation source are lost.

From overvoltage theory, losing a considerable load together with a supply line produces an overvoltage. As demand falls, and according to the droop characteristic, the frequency varies: ideally 60 Hz during the first 2.5 s, but once the load is disconnected, generation exceeds demand and the frequency rises (it falls in the opposite case). Fig. 21 shows the phase-to-ground voltage measured with an RMS block downstream of the three-phase breaker: the overvoltage caused by the load disconnection at 2.5 s is visible, followed by stabilization after 6 s; reconnecting the load produces a short-lived transient overvoltage.

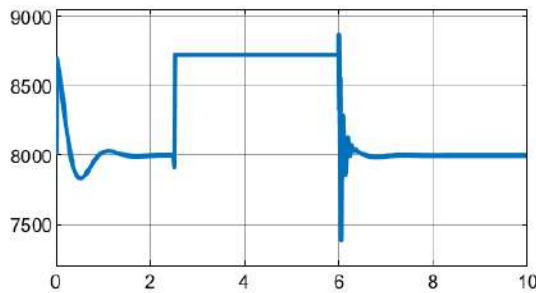


Fig. 21. Voltage downstream of the breaker during the disconnection

At the generator output, the signal is steady before disconnection, but once the load is removed both the amplitude and the frequency vary, departing from the ideal 60 Hz. This variation depends on the action of both the exciter and the governor.

As load is lost, the torque exceeds what the current load requires, which raises speed and frequency. If left uncorrected, this could lead the system to collapse, and this is where the governor acts. In general terms, the governor regulates the generator speed using frequency sensors: when the frequency departs from its reference, the governor acts to compensate the change by delivering more or less mechanical power to the prime mover. Fig. 22 shows the governor response to the load loss: the delivered power oscillates until it finds the appropriate value.

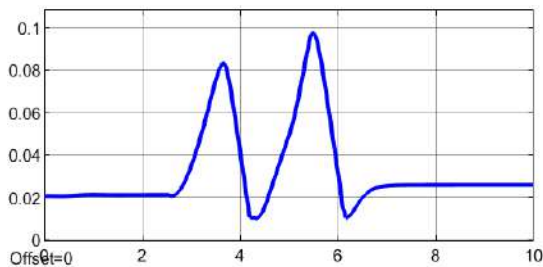


Fig. 22. Mechanical power delivered by the governor

During the first 2.5 s the governor delivers a constant power, the nominal value needed to keep the system at 60 Hz. After 2.5 s this power oscillates and does not fully settle: on detecting the load loss, the governor first supplies more power to probe the system response, then reduces it once it

finds that more power does not correct the frequency, and oscillates again until the signal settles. This reflects a PID-type control, characterized by initial oscillations that depend on its gain constants. Because the controller was left with its default constants, it is slow to reach the reference; in addition, the breaker recloses after a relatively short time, leaving little time to settle. Separate tests show that stabilization takes longer than expected, which highlights the importance of properly tuning the governor.

After reconnection, the power settles at a constant value higher than the initial one. Three operating stages can therefore be identified: a stable state, a load-loss state, and a load-recovery state, each with distinct consequences. Because the recovered load is perceived as a new (and larger) load, the governor delivers slightly more power to keep the system stable, and the mechanical power stops fluctuating. In short, the mechanical power increases with the load in order to keep the frequency stable.

Frequency and speed are directly related. As established in the motor simulations, synchronous machines keep a constant speed; in the generator, keeping the frequency constant likewise keeps the speed constant, as shown in Fig. 23. When the load is disconnected, the speed fluctuates—because the governor, which regulates speed, does not settle between 2.5 and 6 s—and oscillates until the governor power becomes constant again.

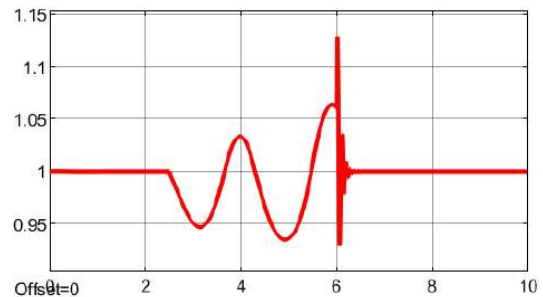


Fig. 23. Generator speed in per unit

Speed is also controlled through the field (excitation) voltage. By analogy with DC machines, a higher field voltage increases the field current and the magnetic flux, and an increase in flux reduces the speed—a relationship that applies here specifically because the source is islanded after the breaker isolates it. There are thus two speed-control paths: the governor and the excitation. The excitation voltage during the simulation is shown in Fig. 24.

The excitation fluctuates between positive and negative values. During start-up the excitation voltage is negative: the machine behaves as a motor because the generator has not yet started and only mechanical power is supplied to the prime mover; the low excitation also corresponds to a higher speed. The excitation then becomes constant, and the largest fluctuations begin exactly when the load is disconnected.

When the load is disconnected, the frequency and speed rise; the exciter detects this and increases the field excitation to

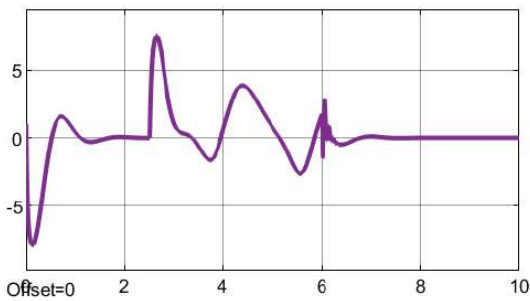


Fig. 24. Excitation voltage

reduce the speed. As Fig. 25 shows, the excitation increases at disconnection to counteract the speed rise, but the correction is excessive, so the governor then increases its power delivery to recover the resulting speed drop. When the governor delivers too much power, the speed exceeds nominal and the exciter acts again. The governor and exciter thus alternate to keep speed and frequency constant: each block depends on and optimizes the action of the other.

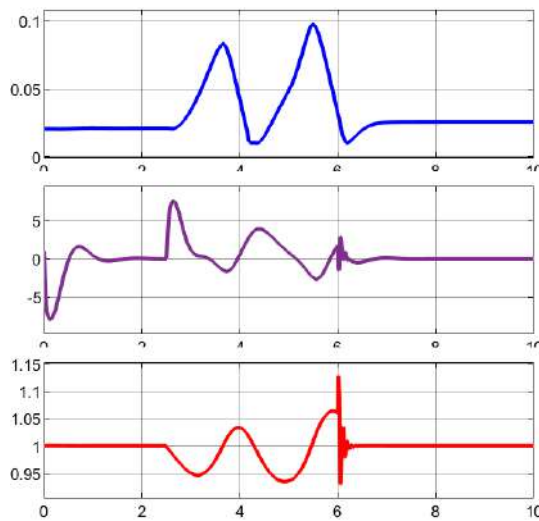


Fig. 25. Comparison of mechanical power, excitation voltage, and generator speed

After the load is reconnected, the system perceives a new, larger load; to keep the speed constant under this condition, the governor power and the excitation level are both higher than at the start, reaching a balance between the two blocks. A final confirmation is the load angle in Fig. 26.

Initially the machine operates as a motor (negative load angle) and then settles as a generator (positive, constant angle) during the first 2.5 s. At disconnection, the angle fluctuates—behaving like the power—and at reconnection it rises, because the system recognizes the larger load and must adapt all its parameters. A larger load angle implies larger delivered power.

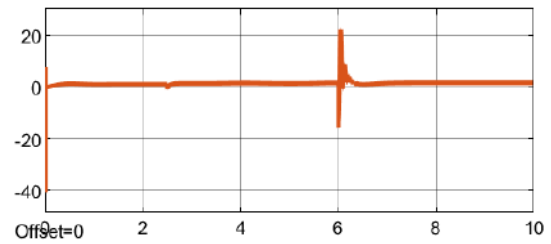


Fig. 26. Load angle, exercise 3

V. CONCLUSIONS

The practice clarified the operation and importance of synchronous machines in electrical engineering. Two motor studies (load change and field-current change) and one generator study (line-opening maneuver in a power system) were carried out.

In the load-change study, the key observations were the response of speed, torque, and load angle. A sudden load reduction leaves an excess of torque, so the shaft speed tends to rise; because the synchronous motor keeps a constant speed, it reduces the torque to bring the speed back down. The load angle follows the power: when the angle increased, both the delivered power and the torque increased, and vice versa. This was confirmed across the three scenarios—nominal, load halved at 0.2 s, and load increased at 1 s. A load change in a synchronous motor therefore has a direct effect on its load angle and torque.

In the field-current study, the combined scenario showed that, at the nominal state, the machine reaches a specific load angle, a high power factor, and positive reactive power (inductive regime). When the field voltage is reduced, the load angle increases, the power factor decreases slightly, and the reactive power becomes negative (capacitive regime). Two conclusions follow: first, with constant load and varying field voltage, the load angle is the first quantity to change; second, and more importantly, depending on its operating conditions a synchronous motor can replace capacitor banks to adjust the system power factor—absorbing reactive power here, or supplying it when the field voltage is increased. A small transient speed dip accompanied the field reduction and was then damped by the machine torque; this is a dynamic effect, not a steady-state proportionality, since the synchronous motor always returns to synchronous speed.

In the generator study, opening the line produced a momentary overvoltage and so did the reconnection. The governor and the exciter were both found to regulate the generator speed, which translates into controlling the system frequency. It is important to note that a generator is the dual of a motor: whereas reducing the field current of the motor lowered its speed, reducing the field current of the islanded generator raises its speed, and increasing the excitation lowers it; when the excitation is negative, the machine behaves as a motor.

When the load is disconnected, the system shows a clear droop case: generation exceeds demand and the frequency

rises. Real power systems use frequency sensors that trigger the speed-control equipment—here the governor and the exciter—when limits are exceeded. The exciter acts first, raising the field voltage to reduce the speed; the reduction is excessive, so the governor delivers more mechanical power to raise the speed again; this in turn overshoots, and the two blocks alternate without reaching stability in this scenario, mainly because losing the line also removes the grid reference.

Once the load is reconnected, the analyzed quantities begin to stabilize, but several values—torque, load angle, field voltage, and governor power—settle above their initial levels. With the grid reconnected, the system again has a frequency reference, and because the reconnected load is perceived as larger, the governor increases the mechanical power while the exciter reduces the field voltage until both blocks reach equilibrium; the torque also has to increase for the new load.

Overall, the practice clarified how these machines and control elements operate and how they relate to other subjects—primarily Automatic Control, since elements such as the governor use PID-based control, and Optimization, given the real-world problems of generation dispatch and unit commitment that, although not analyzed here, are central to power systems. The two most common synchronous machines—motor and generator—were studied, showing that a single machine can operate in both modes, each with its own advantages and limitations: a synchronous motor can improve the power factor in an industrial plant, while the generator plays a central role in maintaining system stability.

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